I. Project Overview

Students are required to design and implement a database system based on an approved case study. The project must demonstrate proper database design, normalization, and correct SQL implementation within a working application.

II. Minimum Technical Requirement

The project must use a relational database management system (RDBMS), with Microsoft SQL Server (MSSQL Server) as the minimum requirement. The application must successfully connect and operate with the chosen DBMS, demonstrating proper database configuration and reliable connectivity throughout the system.

III. Database Design Requirements

A. **Entity Relationship Diagram (ERD)**

The ERD must clearly show:

- All entities (tables)
- Attributes (columns)
- Primary Keys (PK)
- Foreign Keys (FK)
- Relationships (1:1, 1:M, M:N)
- Proper cardinality

The database must be normalized up to Third Normal Form (3NF).

B. Data Dictionary

A complete Data Dictionary must be provided for all tables.

Each table must include:

- Table Name
- Column Name
- Data Type
- Length/Size
- Constraints (PK, FK, NOT NULL, UNIQUE, DEFAULT, etc.)
- Description of the field

The Data Dictionary must clearly explain the purpose of each column.

C. Core Data Requirement

Each core table in the database must contain at least 20 records to effectively demonstrate the main processes of the application. The data should be meaningful and consistent with the system's business logic, ensuring that all SQL operations and transactions can be properly tested and validated within the application.

IV. Project Application Requirements

The system must demonstrate actual SQL usage using MSSQL Server.

A. Database Connection

The application must include a working connection to MSSQL Server.
The connection setup must include:

- Server Name
- Database Name
- Authentication Type (Windows or SQL Authentication)
- Username (if applicable)
- Password (may be hidden in documentation)

B. Required SQL Operations

The system must properly implement the following operations:

1. SELECT (Data Retrieval)
    - Do not use SELECT *
    - Specify required columns only
    - Use WHERE conditions properly
    - Use JOIN when retrieving related data
2. INSERT (Data Insertion)
    - Always specify column names
    - Insert validated data
3. UPDATE (Data Modification)
    - Always include a WHERE condition
    - Prevent unintentional updates of all records
4. DELETE (Data Removal)
    - Always include a WHERE condition
    - Ensure controlled deletion of records

C. SQL Transactions

The system must implement transactions when performing related operations.
The following must be demonstrated:

```
BEGIN TRANSACTION
COMMIT
ROLLBACK
```

Transactions must be used to maintain data integrity. If an error occurs during a multi-step process, the system must execute a ROLLBACK.

V. General Guidelines

- Use proper naming conventions for tables and columns.
- Use appropriate data types.
- Enforce referential integrity using foreign key constraints.
- Apply normalization principles to prevent redundancy and data anomalies.
- Follow SQL best practices for readability and security.

Database Evaluation Rubric											
#	Category	Criteria	Excellent		Good		Fair		Poor		Points
1	**Database Design**	ERD – Complete entities, attributes, PK, FK, relationships, normalized to 3NF	All correct and normalized	24	Minor missing elements	19	Several missing/incorrect	14	Incomplete or incorrect	1	24
2	**Database Design**	Data Dictionary – Columns, data types, constraints, descriptions	All tables fully documented	14	Minor missing descriptions/constraints	11	Missing several fields or errors	8	Incomplete or inaccurate	1	14
3	**Project Application**	RDBMS Implementation – Works with chosen RDBMS, MSSQL minimum	Fully working connection and config	10	Minor issues, still functional	8	Partial functionality	5	Not working / not implemented	1	10
4	**Project Application**	Core Table Data – Each core table ≥ 20 meaningful, consistent records	All core tables with ≥20 valid records	5	Minor inconsistencies	4	Some tables <20 records	3	Most tables missing / invalid data	1	5
5	**SQL Operations**	SELECT – Proper columns, WHERE, JOIN, no SELECT *	Fully correct, best practices	5	Minor issues	4	Partially correct	3	Incorrect or missing	1	5
6	**SQL Operations**	INSERT – Correct structure, column names, valid data	Fully correct	5	Minor errors	4	Partially correct	3	Incorrect or missing	1	5
7	**SQL Operations**	UPDATE – Proper use of WHERE condition	Fully correct	5	Minor issues	4	Partial correctness	3	Incorrect / missing	1	5
8	**SQL Operations**	DELETE – Proper use of WHERE condition	Fully correct	5	Minor issues	4	Partial correctness	3	Incorrect / missing	1	5
9	**SQL Transactions**	BEGIN TRANSACTION, COMMIT, ROLLBACK correctly implemented	Fully correct	10	Minor issues	8	Partial implementation	5	Not implemented	1	10
10	**Code Quality & Best Practices**	Naming conventions, correct data types, referential integrity, clean/readable SQL/code	Excellent / fully organized	17	Good / minor improvements	14	Fair / needs improvement	10	Poor / messy / incomplete	1	17
									Total		**100**